

PART A — QUESTIONS

AF12-001

Question:

Solve: $14x + 8 = 92$

- A) 5
- B) 6
- C) 7
- D) 8

AF12-002

Question:

If $12n - 5 = 91$, what is n ?

- A) 6
- B) 7
- C) 8
- D) 9

AF12-003

Question:

Simplify: $5(3x + 4) - 7x$

- A) $8x + 20$
- B) $15x + 20$
- C) $8x + 4$
- D) $22x + 20$

AF12-004

Question:

A service charges \$20 per hour plus a \$50 flat fee. Which expression represents the total cost for h hours?

- A) $20h + 50$
- B) $50h + 20$
- C) $70h$
- D) $50 - 20h$

AF12-005

Question:

If $f(x) = 9x - 10$, what is $f(6)$?

- A) 34
- B) 44
- C) 54
- D) 64

AF12-006

Question:

Which ordered pair satisfies the equation $y = 7x - 3$?

- A) (1, 5)
- B) (2, 11)
- C) (3, 19)
- D) (4, 26)

AF12-007

Question:

Solve: $13(x - 2) = 78$

A) 6

B) 7

C) 8

D) 9

AF12-008

Question:

A pattern begins: 10, 21, 32, 43, ... What is the next number?

A) 50

B) 52

C) 53

D) 54

AF12-009

Question:

Solve: $x/20 = 3$

A) 23

B) 40

C) 60

D) 80

AF12-010

Question:

A line has a slope of -9 and crosses the y-axis at 4. Which equation represents the line?

A) $y = 4x - 9$

B) $y = -9x + 4$

C) $y = 9x + 4$

D) $y = -4x + 9$

AF12-011

Question:

If 7 boxes contain 126 parts, how many parts are in 5 boxes at the same rate?

A) 72

B) 90

C) 105

D) 108

AF12-012

Question:

Solve: $80 - 12x = 32$

A) 3

B) 4

C) 5

D) 6

AF12-013

Question:

Which expression is equivalent to $40r - 17r - 6$?

- A) $23r - 6$
- B) $57r - 6$
- C) $23r + 6$
- D) $40r - 23$

AF12-014

Question:

A tank starts with 360 gallons and loses 18 gallons each hour. Which equation gives the amount A left after h hours?

- A) $A = 360h - 18$
- B) $A = 18h + 360$
- C) $A = 360 - 18h$
- D) $A = 18h - 360$

AF12-015

Question:

If $y = -11x + 100$, what is y when $x = 8$?

- A) 12
- B) 18
- C) 88
- D) 188

AF12-016

Question:

Solve: $18x - 15 = 11x + 48$

- A) 7
- B) 8
- C) 9
- D) 10

AF12-017

Question:

Which value of x makes $10x + 5$ greater than 75?

- A) 6
- B) 7
- C) 8
- D) 5

AF12-018

Question:

A table shows a linear relationship.

x: 0, 1, 2, 3

y: 3, 15, 27, 39

What is the rule?

- A) $y = 3x + 12$
- B) $y = 12x + 3$
- C) $y = 15x - 12$
- D) $y = x + 3$

AF12-019

Question:

Solve: $9(x + 4) = 6x + 60$

- A) 6
- B) 7
- C) 8
- D) 9

AF12-020

Question:

If $a = -4$ and $b = 10$, what is $4a + 2b$?

- A) -36
- B) -4
- C) 4
- D) 36

AF12-021

Question:

Which point is located in Quadrant IV?

- A) (8, -6)
- B) (-8, 6)
- C) (-8, -6)
- D) (8, 6)

AF12-022

Question:

A worker checks 120 items in 10 minutes. At the same rate, how many items can the worker check in 7 minutes?

- A) 72
- B) 80
- C) 84
- D) 90

AF12-023

Question:

Solve for p : $L = 15p + 45$

- A) $p = L - 45$
- B) $p = (L - 45)/15$
- C) $p = 15(L - 45)$
- D) $p = L/15 + 45$

AF12-024

Question:

What is the slope of a line that goes through (4, 10) and (10, 46)?

- A) 4
- B) 5
- C) 6
- D) 36

AF12-025

Question:

Simplify: $28x + 9 - 19x - 21$

- A) $9x - 12$
- B) $47x - 12$
- C) $9x + 30$
- D) $47x + 30$

AF12-026

Question:

If $g(x) = x^2 - 7x$, what is $g(11)$?

- A) 33
- B) 44
- C) 77
- D) 121

AF12-027

Question:

A sequence follows the rule “add 8, then multiply by 3.” If the first term is 4, what is the second term?

- A) 12
- B) 20
- C) 24
- D) 36

AF12-028

Question:

Solve: $12(x - 3) + 9 = 81$

- A) 7
- B) 8
- C) 9
- D) 10

AF12-029

Question:

A line crosses the y-axis at -11 and rises 5 units for every 1 unit to the right. Which equation represents the line?

- A) $y = -11x + 5$
- B) $y = 5x - 11$
- C) $y = -5x - 11$
- D) $y = 11x + 5$

AF12-030

Question:

If $14/18 = x/54$, what is x ?

- A) 28
- B) 36
- C) 40
- D) 42

AF12-031

Question:

Solve: $-14x + 28 = -84$

- A) -8
- B) -4
- C) 4
- D) 8

AF12-032

Question:

A warehouse starts with 1,200 labels. Each order uses 60 labels. Which expression gives the number of labels left after o orders?

- A) $1,200 + 60o$
- B) $1,200 - 60o$
- C) $60o - 1,200$
- D) $1,200 \div 60o$

AF12-033

Question:

Which equation has $x = 17$ as its solution?

- A) $2x + 11 = 44$
- B) $4x - 13 = 55$
- C) $x/17 + 5 = 7$
- D) $3x - 8 = 40$

PART B — ANSWER KEY + EXPLANATIONS

AF12-001 — Correct: B — Explanation:

Solve $14x + 8 = 92$. Subtract 8 from both sides: $14x = 84$. Divide by 14: $x = 6$. The most common mistake is subtracting 8 correctly but forgetting to divide by 14.

AF12-002 — Correct: C — Explanation:

Start with $12n - 5 = 91$. Add 5 to both sides: $12n = 96$. Divide by 12: $n = 8$. A common mistake is subtracting 5 instead of adding it.

AF12-003 — Correct: A — Explanation:

Distribute first: $5(3x + 4) = 15x + 20$. Then subtract $7x$: $15x - 7x + 20 = 8x + 20$. A common mistake is forgetting to multiply 4 by 5.

AF12-004 — Correct: A — Explanation:

The service charges \$20 for each hour, so that part is $20h$. The \$50 flat fee is added one time. The total cost is $20h + 50$. A common mistake is treating the flat fee as the hourly rate.

AF12-005 — Correct: B — Explanation:

Substitute 6 for x : $f(6) = 9(6) - 10 = 54 - 10 = 44$. A common mistake is adding 10 instead of subtracting it.

AF12-006 — Correct: B — Explanation:

Test the choices in $y = 7x - 3$. For $(2, 11)$, $7(2) - 3 = 14 - 3 = 11$, which matches the y -value. A common mistake is choosing a point that looks close without checking the equation.

AF12-007 — Correct: C — Explanation:

Solve $13(x - 2) = 78$. Divide both sides by 13: $x - 2 = 6$. Add 2: $x = 8$. A common mistake is adding 2 before undoing the multiplication by 13.

AF12-008 — Correct: D — Explanation:

The pattern increases by 11 each time: 10, 21, 32, 43. Add 11 to 43: 54. A common mistake is assuming the pattern increases by 10 because the first term is 10.

AF12-009 — Correct: C — Explanation:

$x/20 = 3$ means x divided by 20 equals 3. Multiply both sides by 20: $x = 60$. A common mistake is adding 20 and 3 instead of multiplying.

AF12-010 — Correct: B — Explanation:

Slope-intercept form is $y = mx + b$. The slope is -9 and the y-intercept is 4, so the equation is $y = -9x + 4$. A common mistake is switching the slope and y-intercept.

AF12-011 — Correct: B — Explanation:

Find the unit rate: $126 \text{ parts} \div 7 \text{ boxes} = 18 \text{ parts per box}$. For 5 boxes: $18 \times 5 = 90$. A common mistake is multiplying 126 by 5 without finding the rate first.

AF12-012 — Correct: B — Explanation:

Solve $80 - 12x = 32$. Subtract 80 from both sides: $-12x = -48$. Divide by -12: $x = 4$. A common mistake is losing the negative sign after subtracting 80.

AF12-013 — Correct: A — Explanation:

Combine like terms: $40r - 17r = 23r$. The constant -6 stays the same. The expression becomes $23r - 6$. A common mistake is combining the constant with the variable terms.

AF12-014 — Correct: C — Explanation:

The tank starts with 360 gallons and loses 18 gallons each hour. After h hours, $18h$ gallons are lost. The amount left is $A = 360 - 18h$. A common mistake is adding $18h$ even though the tank is losing water.

AF12-015 — Correct: A — Explanation:

Substitute $x = 8$ into $y = -11x + 100$: $y = -11(8) + 100 = -88 + 100 = 12$. A common mistake is treating $-11(8)$ as positive 88.

AF12-016 — Correct: C — Explanation:

Solve $18x - 15 = 11x + 48$. Subtract $11x$: $7x - 15 = 48$. Add 15: $7x = 63$. Divide by 7: $x = 9$. A common mistake is adding 15 to only one side.

AF12-017 — Correct: C — Explanation:

Solve $10x + 5 > 75$. Subtract 5: $10x > 70$. Divide by 10: $x > 7$. The only listed value greater than 7 is 8. A common mistake is choosing 7, but $10(7) + 5 = 75$, not greater than 75.

AF12-018 — Correct: B — Explanation:

The y-values increase by 12 each time x increases by 1, so the slope is 12. When $x = 0$, $y = 3$, so the y-intercept is 3. The rule is $y = 12x + 3$. A common mistake is using 3 as the slope because it is the first y-value.

AF12-019 — Correct: C — Explanation:

Distribute: $9x + 36 = 6x + 60$. Subtract $6x$: $3x + 36 = 60$. Subtract 36: $3x = 24$. Divide by 3: $x = 8$. A common mistake is forgetting to multiply 4 by 9.

AF12-020 — Correct: C — Explanation:

Substitute $a = -4$ and $b = 10$: $4a + 2b = 4(-4) + 2(10) = -16 + 20 = 4$. A common mistake is treating $4(-4)$ as positive 16.

AF12-021 — Correct: A — Explanation:

Quadrant IV has a positive x-coordinate and a negative y-coordinate. The point $(8, -6)$ fits this description. A common mistake is choosing $(-8, 6)$, which is Quadrant II.

AF12-022 — Correct: C — Explanation:

120 items in 10 minutes means $120 \div 10 = 12$ items per minute. In 7 minutes: $12 \times 7 = 84$. A common mistake is multiplying 120 by 7 without finding the rate first.

AF12-023 — Correct: B — Explanation:

Start with $L = 15p + 45$. Subtract 45 from both sides: $L - 45 = 15p$. Divide by 15: $p = (L - 45)/15$. A common mistake is dividing L by 15 first and then subtracting 45.

AF12-024 — Correct: C — Explanation:

Slope equals change in y divided by change in x . From $(4, 10)$ to $(10, 46)$, y increases by 36 and x increases by 6. The slope is $36 \div 6 = 6$. A common mistake is using 36 as the slope without dividing by the change in x .

AF12-025 — Correct: A — Explanation:

Combine like terms: $28x - 19x = 9x$, and $9 - 21 = -12$. The simplified expression is $9x - 12$. A common mistake is writing $9x + 12$ because the negative sign before 21 is missed.

AF12-026 — Correct: B — Explanation:

Substitute 11 for x : $g(11) = 11^2 - 7(11) = 121 - 77 = 44$. A common mistake is calculating 11^2 as 22 instead of 121.

AF12-027 — Correct: D — Explanation:

Start with 4. Add 8: 12. Then multiply by 3: 36. A common mistake is multiplying first and then adding 8.

AF12-028 — Correct: C — Explanation:

Solve $12(x - 3) + 9 = 81$. Distribute: $12x - 36 + 9 = 81$. Combine: $12x - 27 = 81$. Add 27: $12x = 108$. Divide by 12: $x = 9$. A common mistake is forgetting to combine -36 and $+9$.

AF12-029 — Correct: B — Explanation:

The line rises 5 units for every 1 unit to the right, so the slope is 5. It crosses the y -axis at -11 , so the equation is $y = 5x - 11$. A common mistake is switching the slope and intercept.

AF12-030 — Correct: D — Explanation:

Use the proportion $14/18 = x/54$. Since $18 \times 3 = 54$, multiply 14 by 3: $x = 42$. A common mistake is adding 36 to the numerator because 18 becomes 54.

AF12-031 — Correct: D — Explanation:

Solve $-14x + 28 = -84$. Subtract 28 from both sides: $-14x = -112$. Divide by -14 : $x = 8$. A common mistake is giving -8 because both sides contain negative numbers.

AF12-032 — Correct: B — Explanation:

The warehouse starts with 1,200 labels. Each order uses 60 labels, so o orders use $60o$ labels. The number left is $1,200 - 60o$. A common mistake is adding $60o$ even though labels are being used.

AF12-033 — Correct: B — Explanation:

Test $x = 17$. In choice B, $4(17) - 13 = 68 - 13 = 55$, so the equation is true. A common mistake is choosing a nearby equation without checking both sides exactly.